Mohamed Ali Trabelsi



Backend Engineer

Tunis, Tunisia | Open to relocation in Europe and remote work
trabelsi.mohamedali.dev@gmail.com | +216 28 751 504
linkedin.com/in/mohamedalitrabelsi | github.com/MohamedAliTrabelsiSE
Portfolio: mohamedali-trabelsi.web.app

SUMMARY

Backend engineer with two years in industry, working mostly in TypeScript and NestJS. I build REST APIs and services on PostgreSQL and MongoDB for web and mobile products. I have also worked with RabbitMQ microservices, real-time features and LLM integrations. Looking for a backend role in Europe.

SKILLS

LanguagesTypeScript, JavaScript, Java, Python
BackendNestJS, Node.js, Express, Spring Boot, REST APIs, microservices, RabbitMQ, WebSocket, Socket.io, JWT auth
DatabasesPostgreSQL, MongoDB, Supabase
DevOpsDocker, GitHub Actions, CI/CD, DigitalOcean, Vercel, Firebase, Git
TestingJest, Playwright (e2e)
Frontend and mobile React, React Native, Expo, Next.js, Tailwind CSS, Angular
AILLM integration, OpenAI Realtime API, function calling, RAG, prompt engineering

EXPERIENCE

Full-Stack Engineer Sep 2024 – Present
IT Strategix, Tunis, Tunisia

- Build and maintain REST APIs and backend services in NestJS and TypeScript for web and mobile products, on PostgreSQL and MongoDB.
- Cut API response times by 35%.
- One of two backend engineers on the team, with the team lead. I deliver 2–3 tickets a day, depending on complexity, in 2–3 week sprints.
- Take part in architecture decisions and production deployments across several client projects (details confidential).
- Write Jest tests and maintain CI/CD pipelines in GitHub Actions.
- Build features on the React Native (Expo) apps that use these APIs.
- Work with cross-functional teams to pin down requirements for new features.

Stack: NestJS, TypeScript, PostgreSQL, MongoDB, React Native, Expo, Python, Jest, GitHub Actions

Full-Stack Developer Intern Jul – Aug 2024 and Jun – Jul 2025
NAXXUM GROUP, Tunis, Tunisia

- Built backend services and responsive interfaces for an online teaching platform, including real-time student monitoring.
- Worked on backend scalability and frontend performance of an educational platform, and added AI-powered recommendations.

Stack: Node.js, Express, React, MongoDB, REST APIs

End-of-Studies Intern Feb – Jun 2023
SIGA, Tunis, Tunisia

- Built a task-tracking web application with Spring Boot, Angular and PostgreSQL.
- Wrote the REST APIs and the user management module used by internal teams.

Stack: Java, Spring Boot, Angular, PostgreSQL

PROJECTS

Realtime voice agent github.com/MohamedAliTrabelsiSE/realtime-voice-agent
AI voice receptionist. Speech-to-speech appointment booking over WebSocket.

- NestJS backend that streams microphone audio to the OpenAI Realtime API. The model books, finds and cancels appointments through typed tools.
- Low-latency speech-to-speech loop with barge-in. Function calling with Zod schemas and structured errors.
- Session tokens, browser test lab, CI and e2e tests. A mock mode runs the full stack offline without an API key.

Stack: NestJS, TypeScript, OpenAI Realtime, WebSocket, Zod, Jest, Docker

Delivery route tracker github.com/MohamedAliTrabelsiSE/delivery-route-tracker
Courier map on a phone with live tracking and no Google Maps key.

- Expo app and NestJS API for a driver tour and a customer tracking code. Routing and geocoding use OSRM and Nominatim on OpenStreetMap.
- A WebSocket stream moves the driver, updates the ETA and redraws the road still left. Geofences and turn-by-turn.
- Route optimize, reverse and recalculate on the driver tour. Shared geo package, mock providers and Playwright e2e tests.

Stack: NestJS, Expo, React Native, TypeScript, WebSocket, OSRM, OpenStreetMap

Smart delivery mobile app (CONFIRMIA) Capstone, pre-production, confidential
Logistics app on microservices with three user roles, live tracking and an AI voice assistant.

- Microservices backend in NestJS. Services talk through RabbitMQ; data is in Supabase and MongoDB.
- Three roles in a single React Native / Expo codebase. Real-time tracking with Google Maps and QR-code delivery confirmation.
- Voice assistant in Tunisian Arabic (Darija) on the OpenAI Realtime model. Deployed to pre-production on DigitalOcean; mobile builds with Expo EAS.

Stack: NestJS, React Native, Expo, RabbitMQ, Supabase, MongoDB, Google Maps, OpenAI Realtime, DigitalOcean

GoTogether Live demo: gotogether-green.vercel.app
Group travel planner: groups, trips, expenses, chat and a community feed.

- Groups with invites and roles, trips with an itinerary on a map, expense splitting, group and trip chat, community posts.
- JWT sessions, email verification, password reset and TOTP 2FA.

Stack: Next.js, React, Tailwind CSS, MongoDB, JWT, Socket.io

AI-powered education platform
Mobile tutoring app where LLaMA matches students with teachers.

- Expo app and NestJS API on PostgreSQL. LLaMA personalizes tutoring and matches learners with teachers from their needs; the model stays behind the API.

Stack: Expo, React Native, NestJS, PostgreSQL, TypeScript, LLaMA

AI-powered fitness coaching app
Beginner iOS coach with Gemini and Mistral behind the API.

- Swift client, NestJS API and MongoDB. Gemini and Mistral generate training programs from user goals and are isolated in the API.

Stack: Swift, NestJS, MongoDB, Gemini, Mistral AI

EDUCATION

Engineering Degree in Software Engineering 2023 – 2026
ESPRIT, Tunis, Tunisia

Bachelor's Degree in Computer Science 2020 – 2023
Faculty of Sciences of Tunis (FST), Tunis, Tunisia

LANGUAGES AND ACTIVITIES

Languages: Arabic (native), English (C1), French (B2)
Activities: Tunisian Red Crescent; IEEE Student Branch, FST; Engineers Spark, FST; Hult Prize, ESSEC Tunis